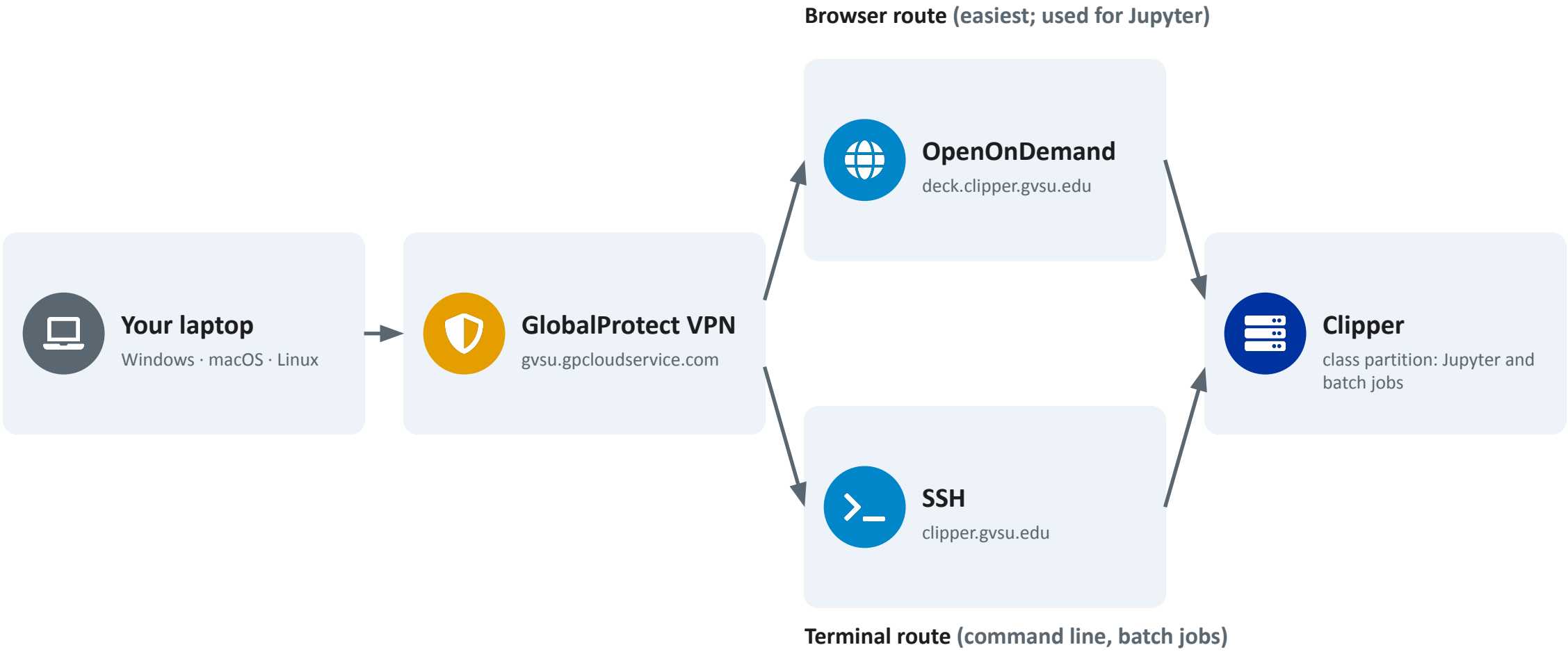


Connecting to the GVSU Clipper HPC Cluster

CIS 635 · VPN, OpenOnDemand, SSH and Jupyter — with troubleshooting

Two ways into Clipper, one gate: the VPN



On student Wi-Fi or off campus, Clipper is unreachable without the VPN. With the VPN on, SSH needs no Duo prompt.

Before you start



GVSU account

Username is the part before @mail.gvsu.edu, plus your network password.



MFA enrolled

The VPN requires it. Enroll at my.gvsu.edu (Okta Verify recommended).



Admin rights

Needed on your laptop to install the GlobalProtect client.



An SSH client

Terminal on macOS/Linux; PowerShell on Windows 10 (1809+) and 11.

First day? Access syncs to Okta over a few hours. If OpenOnDemand refuses you, wait and retry before reporting a problem.

Step 1 · Install and connect the GlobalProtect VPN

Portal address (all platforms)

gvsu.gpcloudservice.com

Windows

1. Download from the GVSU BYOD guide (ARM laptops: “Surface” build)
2. Next → approve UAC → Close
3. Restart, click the globe icon in the tray
4. Enter portal → Connect → sign in + MFA

macOS

1. Download the .pkg from the BYOD guide
2. Continue → Install → Mac password → Close
3. Globe icon in the menu bar
4. Enter portal → Connect

Linux (Ubuntu)

1. Download and extract the .tgz
2. Run gp_install.sh, enter password
3. Enter portal → Connect

Pass: GlobalProtect shows Connected and a gateway (e.g., US Central). On campus it may say “internal corporate network”; on student Wi-Fi use Change Gateway → US Central.

Step 2 · Test the OpenOnDemand web login

1. With the VPN connected, open deck.clipper.gvsu.edu
2. Sign in on the Okta page with your GVSU credentials
3. Dashboard appears: Files, Jobs, Clusters, Interactive Apps
4. Clusters → Clipper Shell Access; type hostname and press Enter
5. Files → Home Directory: where you upload data and notebooks

`deck.clipper.gvsu.edu`

The web portal runs the shell, file browser and Jupyter in your browser — nothing to install beyond the VPN.

Pass: the dashboard loads and your home directory opens. Screenshot the dashboard for the check-in.

Step 3 · Test the SSH connection

1. With the VPN connected, open Terminal or PowerShell:

```
ssh username@clipper.gvsu.edu
```

1. First time: confirm host is clipper.gvsu.edu, type yes
2. Enter your GVSU password (nothing shows as you type)
3. With the VPN on, there is no Duo prompt
4. Type exit when done

2. Run these checks on the login node:

```
whoami                # GVSU username
hostname              # login node name
pwd                   # home directory
python3 --version     # system 3.9
module avail python   # newer versions
sinfo -p class        # your partition
```



You land on Port, the shared login node. Never run heavy computation there — use Jupyter or a batch job on the class partition.

Your limits on the class partition

4	32 GB	4
CPU cores	memory	GPU shards (slices of a shared GPU)

- Limits are per student, across all your running jobs; extra jobs wait in the queue.
- A single request above the limit is rejected by Slurm.
- Only the class partition is allowed; any other gives “Invalid account or account/partition combination specified”.
- GPU sessions launched from OpenOnDemand automatically get 4 shards.

Optional · Submit a test batch job

Save as test_job.sh

```
#!/bin/bash
#SBATCH --job-name=cis635_test
#SBATCH --output=output_%j.txt
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=2G
#SBATCH --partition=class

echo "Hello from $(hostname)"
python3 --version
```

1 **`sbatch test_job.sh`**

Submit; prints the job ID

2 **`squeue -u $USER`**

Watch it queue and run

3 **`cat output_<jobid>.txt`**

Read the result

Jupyter · one-time setup of your course environment

Run once from SSH or the OpenOnDemand shell:

```
cd ~
module load python/3.12        # optional; check: module avail python
python3 -m venv cis635
source cis635/bin/activate
python3 -m pip install --upgrade pip
python3 -m pip install ipykernel numpy pandas scikit-learn matplotlib seaborn plotly
python3 -m ipykernel install --user --name cis635 --display-name "Python (cis635)"
deactivate
```

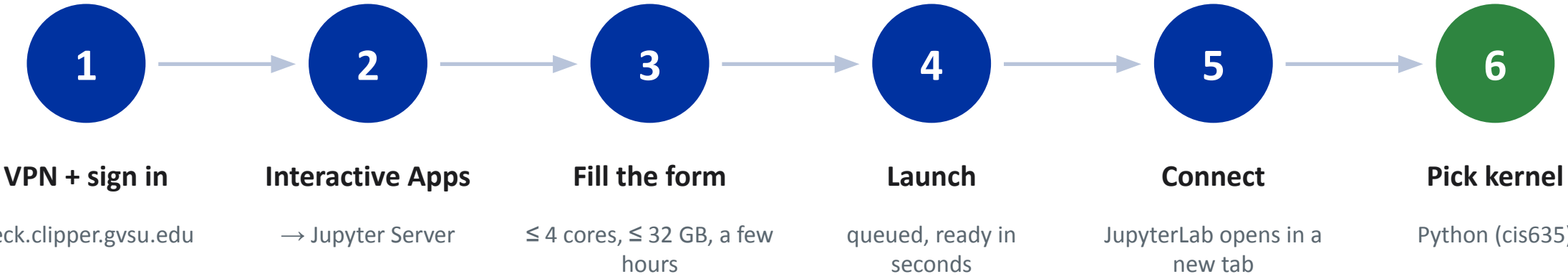


Virtual environments are GVSU's recommended way to manage Python on Clipper (default Python is 3.9).



The ipykernel line makes "Python (cis635)" appear as a kernel in JupyterLab.

Jupyter · launch a session each time



Form tips

- Request only the hours you need (e.g., 2) — the session ends automatically at the time limit.
- Leave GPU off unless an assignment needs it; GPU sessions get 4 shards.
- If every class core is busy, your session waits in the queue.

Jupyter · verify, move files, and shut down

Run in your first notebook cell:

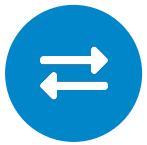
```
import sys, platform, os
import numpy, pandas, sklearn
print("Python:", sys.version.split()[0])
print("Node:", platform.node())
print("Cores:", len(os.sched_getaffinity(0)))
print(pandas.__version__, sklearn.__version__)
```

Node name differs from the login node → you are on a compute node.



Files

Files → Home Directory, JupyterLab's upload button, or scp over the VPN.



From Colab

Drop google.colab lines; read data from your Clipper home directory.



When done

My Interactive Sessions → Delete. Closing the tab leaves cores reserved.

Troubleshooting · getting connected

Symptom	Likely cause	What to do
OpenOnDemand refuses you on day one	Okta not synced yet	Wait a few hours, retry
VPN error right after password	MFA not enrolled	Enroll at my.gvsu.edu, sign out fully, retry
ssh hangs / connection timed out	VPN off (or student Wi-Fi)	Connect GlobalProtect, check Connected
SSH asks for Duo	VPN not connected	Connect the VPN; Duo is bypassed
Permission denied over SSH	Wrong username or password	Use the username only, no @mail.gvsu.edu

Troubleshooting · jobs and Jupyter

Symptom	Likely cause	What to do
Invalid account or account/partition combination	Job sent outside the class partition	Add #SBATCH --partition=class
Job rejected or form error on launch	Over 4 cores / 32 GB / 4 shards	Lower the request
Session stuck at Queued	All class resources in use	Wait, or delete your older sessions
Kernel Python (cis635) missing	Environment not registered	Rerun the ipykernel install line
ModuleNotFoundError in a notebook	Wrong kernel or package missing	Switch kernel; pip install into cis635

Still stuck? Email a screenshot of the error, your OS, and whether the VPN showed Connected.

Connection checklist

- [ ] 1. VPN installed and shows Connected
- [ ] 2. OpenOnDemand dashboard loads (screenshot)
- [ ] 3. SSH works; whoami and hostname print (screenshot)
- [ ] 4. cis635 environment registered as a kernel
- [ ] 5. Jupyter session runs the test cell (screenshot)
- [ ] 6. Session deleted afterwards

GVSU KB articles: Accessing Clipper HPC Cluster · GlobalProtect VPN Overview · BYOD GlobalProtect install · HPC Classroom Usage · Accessing JupyterLab on Clipper · Python Environments on Clipper